

EVENT HORIZON

Remote Software Testing, Defect Review & Release Audit

Project: The RTI Black Hole | CX0107 | Team Tensor Float

Audit Item	Value
Target	https://event-horizon-rust.vercel.app/
Audit date	21 September 2026
Scope	UI/UX, navigation, functional behavior, RTI workflow, AI/routing logic, accessibility, performance, security and deployment
Primary evidence	Publicly reachable deployment root and indexed page metadata
Important limitation	The available remote inspection interface could retrieve the deployment title but could not execute or instrument the client-side UI/API. Therefore unverified items are explicitly marked as such.

Audit conclusion: the public deployment is identifiable and reachable at its root, with the page title *Event Horizon - AI-Assisted RTI Jurisdiction Routing Portal*. However, the inspection channel available in this environment does not expose the runtime DOM, browser console, network trace, or source/API implementation. This report therefore separates confirmed observations from high-risk test findings that require browser/repository execution, rather than presenting unverified defects as facts.

This is a professional QA audit artifact and not a claim that every listed risk is currently present.

1. Executive Summary

The Event Horizon deployment was checked as a live public web target. The root deployment is indexed as an AI-assisted RTI jurisdiction-routing portal, which aligns with the project identity. The available inspection path could not render or interact with the application's client-side runtime, so a true end-to-end browser execution pass could not be completed from this environment.

For a project whose critical value proposition is confidence-aware routing, the highest-risk areas are not only visual. The decisive quality gates are: correct handling of CLEAR / AMBIGUOUS / UNKNOWN, resistance to arbitrary routing when jurisdiction is shared or missing, traceable explanation of why a department was suggested, reliable PDF/text extraction, and consistency between frontend state and backend decision output.

Accordingly, the report prioritizes testable failure modes that could cause harmful or misleading routing, followed by UI/UX, accessibility, security, performance and deployment checks.

Severity language: P0 = release blocker; P1 = critical/high impact; P2 = material defect; P3 = minor defect; P4 = polish.

Area	Audit status	Risk
Deployment reachability	Observed at root via web retrieval	Low
Runtime UI execution	Not instrumentable in this environment	Unknown
Navigation/routes	Not executable end-to-end	High if broken
RTI analyzer	Not executable end-to-end	Critical
CLEAR / AMBIGUOUS / UNKNOWN	Requires black-box or source test	Critical
PDF extraction	Requires browser/API test	High
Accessibility	Requires rendered DOM / browser audit	High
Performance	Requires Lighthouse/browser trace	Medium-High
Security headers/API	Requires HTTP/runtime inspection	High

2. Confirmed Remote Observations

OBS-01 - Deployment root is publicly reachable through the web inspection layer. The public page resolves to a titled page rather than an invalid domain. The retrieved page identifies itself as the Event Horizon AI-assisted RTI jurisdiction-routing portal.

OBS-02 - The root response is client-rendered/otherwise not exposed as readable page text to the available inspector. This is not sufficient evidence of a product bug; it is an inspection constraint. It does mean that accessibility, link integrity and runtime behavior cannot be established from the retrieved text alone.

OBS-03 - Direct inspection of guessed subroutes did not succeed through the available web retrieval interface. This is not evidence that the application's routes are broken. It may reflect tool restrictions, SPA routing, deployment configuration or client-side navigation. Route health therefore remains unverified.

3. Defect Register - Confirmed vs Unverified

ID	Finding	Status	Priority	Required proof
RMT-01	Client-side runtime could not be executed/instrumented from the available inspection channel.	Confirmed limitation	P2	Browser run or repo access
FUN-01	CLEAR / AMBIGUOUS / UNKNOWN behavior not independently verified in the live deployment.	Unverified risk	P0	Black-box test with crafted RTIs

ID	Finding	Status	Priority	Required proof
ALG-01	Shared-jurisdiction requests may incorrectly collapse to one department if backend ignores ambiguity.	High-risk test	P0	Algorithm/source test
ALG-02	Unknown subjects may receive a false department if fallback matching is too permissive.	High-risk test	P0	Algorithm/source test
ALG-03	Routing confidence may be displayed as probability rather than heuristic confidence.	High-risk test	P1	UI + response inspection
ALG-04	Explanation may be generic or inconsistent with the actual evidence used for routing.	High-risk test	P1	Result trace comparison
DOC-01	PDF extraction may fail on scanned, multi-column or malformed RTIs.	High-risk test	P1	Upload corpus
UX-01	User may not understand what to do after an AMBIGUOUS result without clear human-review guidance.	High-risk test	P1	Usability test
A11Y-01	Status states may rely on color alone.	High-risk test	P1	DOM + accessibility audit
SEC-01	Upload/API surface may lack file-type, size, rate-limit or input validation controls.	High-risk test	P1	Network/security test

4. Functional Test Plan - End-to-End

The following black-box cases should be executed against the deployed site. These are the minimum release-gate tests for the hackathon prototype.

ID	Test	Expected result	Severity if failed
F-01	Open root URL	Landing page loads without blank screen or runtime error	P1
F-02	Open Analyze RTI from header/CTA	Analyzer opens and is usable	P0
F-03	Submit empty request	Validation blocks submission with clear message	P1
F-04	Submit whitespace-only request	Validation blocks submission	P1
F-05	Submit normal road-repair RTI	Clear result or justified match is shown	P0
F-06	Submit drainage + water maintenance RTI	Ambiguous result if shared jurisdiction applies	P0
F-07	Submit unsupported subject	UNKNOWN; no department fabricated	P0
F-08	Upload valid text PDF	Text extracted and analyzed	P0
F-09	Upload image/scanned PDF	Either OCR succeeds or actionable extraction error is shown	P1
F-10	Upload invalid file	Rejected safely with user guidance	P1
F-11	Network/API timeout	Graceful error; page remains usable	P1
F-12	Refresh result page	State behavior is defined; no broken/empty route	P2
F-13	Back/forward browser navigation	Navigation state remains coherent	P2
F-14	Repeated analysis	Previous state does not leak into new request	P1
F-15	Long RTI input	No layout break; sensible limit or truncation policy	P1

5. Algorithm / Decision-Engine Audit

Because Event Horizon is a routing product, algorithmic correctness is the most important test domain. The audit must prove not only that the model produces a department when it can, but also that it refuses to make a confident choice when evidence is insufficient.

Case	Input pattern	Expected decision	Failure condition
A1	Strong single-subject match + single jurisdiction	CLEAR	Picks unrelated department or abstains unexpectedly
A2	Two subjects mapped to shared jurisdiction	AMBIGUOUS	Automatically chooses one
A3	Subject absent from Star Map	UNKNOWN	Invents a department
A4	Known subject + wrong district	AMBIGUOUS or UNKNOWN	Ignores district and overconfidently routes
A5	Weak keyword overlap only	UNKNOWN/AMBIGUOUS	Keyword match alone produces CLEAR
A6	Strong semantic similarity but explicit conflicting jurisdiction rule	AMBIGUOUS	Semantic similarity overrides conflict
A7	Empty/malformed extracted text	Validation / UNKNOWN	Routes based on stale or default state
A8	Repeated identical input	Deterministic or acceptably stable	Materially different routing without cause

Case	Input pattern	Expected decision	Failure condition
A9	Case/punctuation/language variation	Same conceptual result	Brittle routing
A10	Adversarial irrelevant text around subject	Core subject preserved	Prompt/injection-like text changes routing

Recommended decision invariants:

- AMBIGUOUS must never silently become CLEAR merely because one candidate has a marginally higher score.
- UNKNOWN must never be converted into a department name through a default or fallback value.
- Routing Confidence must be treated as a configurable MVP heuristic, not a calibrated probability.
- The explanation shown to the user should correspond to actual evidence/signals used by the decision engine.
- A shared jurisdiction rule must have higher precedence than a weak semantic match to one department.
- Thresholds should be centralized so boundary behavior can be regression-tested.

6. UI/UX Audit Checklist

Area	Checks	Priority
Landing	Clear value proposition, obvious Analyze CTA, no jargon wall	P1
Analyzer	Readable form, PDF upload state, validation, examples, clear next step	P0
Progress	Meaningful loading stages; no fake indefinite spinner	P1
CLEAR	Department, subject, confidence, evidence, action are scannable	P0
AMBIGUOUS	Multiple candidates shown neutrally; human review is unmistakable	P0
UNKNOWN	No department shown; verification guidance is prominent	P0
Explainability	User can understand why a decision was made	P1
Star Map	Search/filter works and demo data is clearly labeled	P1
Navigation	No dead links, duplicate pages or inconsistent terminology	P1
Errors	Every failure state has recovery guidance	P1
Mobile	No horizontal overflow; touch targets and CTA remain usable	P1
Visual consistency	One component system, consistent spacing, typography, status badges	P2

7. Accessibility Audit

- Keyboard-only navigation: every interactive control must be reachable and visibly focused.
- Status must be communicated using text and iconography, not color alone.
- Form labels and error messages must be programmatically associated.
- Heading hierarchy should be logical and unique.
- Contrast should meet WCAG AA for normal body text and controls.
- File-upload controls must expose clear instructions and accessible status messages.
- Loading updates should use accessible live-region behavior where appropriate.
- Dialogs/modals must trap focus and return focus to the triggering control.
- Tables and result cards must remain understandable at mobile widths.
- Language switching, if offered, should not produce mixed-language or untranslated labels.

8. Security & Trust Audit

Test	What to verify	Priority
File upload	Allow-list file types; enforce size limit; reject polyglot/malformed files	P0
Input handling	Escape/sanitize user RTI text in rendered output	P0
API	Rate limiting, request size limits, CORS scope, error sanitization	P1
Secrets	No API keys or privileged credentials embedded in frontend bundle	P0
Logging	No RTI contents or sensitive data unnecessarily logged	P1
Headers	HTTPS, CSP where feasible, X-Content-Type-Options, Referrer-Policy, frame protection	P1
Prompt/AI safety	User text cannot override system decision policy or force a department	P0
Traceability	Result can be tied to a concrete analysis request and rule version	P1

9. Performance & Reliability

A browser-level performance audit should capture Core Web Vitals and interaction timings. Because the current remote inspection layer did not provide a Lighthouse trace, the values below are release targets rather than measured scores.

Metric	Target	Test
LCP	<= 2.5 s on a representative mobile profile	Lighthouse
INP	<= 200 ms	Lighthouse / interaction trace
CLS	<= 0.10	Lighthouse
Analyzer submit	Clear feedback within 200 ms of click	Browser trace
API timeout	Bounded and user-visible	Network test
Large PDF	No tab freeze; graceful progress/error	Upload stress test
Mobile rendering	No major jank or layout shift	Device/emulation

10. Deployment & Web Hygiene

Item	Expected	Current remote evidence
HTTPS	Valid TLS deployment	Root deployment reachable; detailed header inspection unavailable
Custom 404	Useful, branded 404 page	Unverified
robots.txt	Present if intended	Unverified
sitemap.xml	Present if multi-page/SEO relevant	Unverified
favicon	Present and correct	Unverified
Open Graph	Title/image/description defined	Unverified
Page titles	Unique, descriptive titles	Root title observed
Error monitoring	Client/server errors observable	Unverified
Build/deploy	Repeatable and pinned dependencies	Repo access required

11. Release Gate

For the hackathon demo, do not treat visual polish as sufficient. The product should pass the following hard gates before being presented as a working AI routing system.

Gate	Must pass?
Analyzer accepts valid text and produces a real backend result	YES
CLEAR case routes only on sufficiently strong evidence	YES
AMBIGUOUS case visibly requests human review	YES
UNKNOWN case does not fabricate a department	YES
Explanation matches actual decision signals	YES
PDF failure is handled gracefully	YES
No secrets in frontend bundle	YES
Mobile layout has no blocking defects	YES
No known runtime errors in browser console	YES
No critical accessibility blocker on the main workflow	YES

12. Final QA Assessment

Current assessment: Not release-verifiable from the public URL alone using the available remote inspection channel.

The deployment identity is confirmed, but a full defect verdict cannot responsibly be issued without execution of the client-side application and access to its actual request/response path or repository. The most consequential validation remains the routing decision engine and the three-state safety behavior.

Recommended next artifact for a definitive audit: the Event Horizon repository (or a ZIP of the deployed source), plus any environment-safe sample jurisdiction rules. With source/runtime access, the remaining unverified cases can be converted into concrete PASS/FAIL defects with file/line references.

Reference deployment: <https://event-horizon-rust.vercel.app/>

Publicly visible page title retrieved during audit: Event Horizon - AI-Assisted RTI Jurisdiction Routing Portal.